

5.1.2 SPRING/MASS SYSTEMS: FREE DAMPED MOTION

The concept of free harmonic motion is somewhat unrealistic, since the motion described by equation (1) assumes that there are no retarding forces acting on the moving mass. Unless the mass is suspended in a perfect vacuum, there will be at least a retarding force due to the surrounding medium. As Figure 5.1.5 shows, the mass could be suspended in a viscous medium or connected to a dashpot damping device.

DE of Free Damped Motion In the study of mechanics, damping forces acting on a body are considered to be proportional to a power of the instantaneous velocity. In particular, we shall assume throughout the subsequent discussion that this force is given by a constant multiple of dx/dt . When no other external forces are represented on the system, it follows from Newton's second law that

$$m \frac{d^2x}{dt^2} = -kx - \beta \frac{dx}{dt} \quad (10)$$

where β is a positive damping constant and the negative sign is a consequence of the fact that the damping force acts in a direction opposite to the motion.

Dividing (10) by the mass m , we find that the differential equation of free damped motion is

$$\frac{d^2x}{dt^2} + \frac{\beta}{m} \frac{dx}{dt} + \frac{k}{m} x = 0 \quad (11)$$

where

$$\lambda = \frac{\beta}{m}, \quad \omega^2 = \frac{k}{m} \quad (12)$$

$$m \frac{d^2x}{dt^2} = -kx - \beta \frac{dx}{dt}$$

$$\frac{\beta}{m} = \lambda$$



FIGURE 5.1.5 Damping devices.



FIGURE 5.1.6 Motion of an overdamped system.



FIGURE 5.1.7 Motion of a critically damped system.



FIGURE 5.1.8 Motion of an underdamped system.

CASE I: $\lambda^2 - \omega^2 > 0$ In this situation the system is said to be **overdamped** because the damping coefficient β is larger when compared to the spring constant k . The corresponding solution of (11) is $x(t) = c_1 e^{m_1 t} + c_2 e^{m_2 t}$ or

$$x(t) = e^{-\lambda t/2} (c_1 e^{\sqrt{\lambda^2 - \omega^2} t} + c_2 e^{-\sqrt{\lambda^2 - \omega^2} t}) \quad (13)$$

This equation represents a smooth and nonoscillatory motion. Figure 5.1.6 shows two possible graphs of $x(t)$.

CASE II: $\lambda^2 - \omega^2 = 0$ The system is said to be **critically damped** because any slight decrease in the damping force would result in oscillatory motion. The general solution of (11) is $x(t) = c_1 t e^{-\lambda t/2} + c_2 e^{-\lambda t/2}$ or

$$x(t) = e^{-\lambda t/2} (c_1 t + c_2) \quad (14)$$

Some graphs of typical motion are given in Figure 5.1.7. Notice that the motion is quite similar to that of an overdamped system. It is also apparent from (14) that the mass can pass through the equilibrium position at most one time.

CASE III: $\lambda^2 - \omega^2 < 0$ In this case the system is said to be **underdamped**, since the damping coefficient is small in comparison to the spring constant. The roots m_1 and m_2 are now complex.

Then the general solution of equation (11) is

$$x(t) = e^{-\lambda t/2} (c_1 \cos(\sqrt{\omega^2 - \lambda^2} t) + c_2 \sin(\sqrt{\omega^2 - \lambda^2} t)) \quad (15)$$

As indicated in Figure 5.1.8, the motion described by (15) is oscillatory, but because of the coefficient $e^{-\lambda t/2}$, the amplitudes of vibration $\rightarrow 0$ as $t \rightarrow \infty$.

25. A force of 2 pounds stretches a spring 1 foot. A mass weighing 3.2 pounds is attached to the spring, and the system is then immersed in a medium that offers a damping force that is numerically equal to 0.4 times the instantaneous velocity.

- Find the equation of motion if the mass is initially released from rest from a point 1 foot above the equilibrium position.
- Express the equation of motion in the form given in (23).
- Find the first time at which the mass passes through the equilibrium position heading upward.

26. After a mass weighing 10 pounds is attached to a 5-foot spring, the spring measures 7 feet. This mass is removed and replaced with another mass that weighs 8 pounds. The entire system is placed in a medium that offers a damping force that is numerically equal to the instantaneous velocity.

- Find the equation of motion if the mass is initially released from a point $\frac{1}{2}$ foot below the equilibrium position with a downward velocity of 1 ft/s.
- Express the equation of motion in the form given in (23).
- Find the times at which the mass passes through the equilibrium position heading downward.
- Graph the equation of motion.

ALTERNATIVE FORM OF $x(t)$ In a manner identical to the procedure used on page 184, we can write any solution

$$x(t) = e^{-\lambda t/2} (c_1 \cos(\sqrt{\omega^2 - \lambda^2} t) + c_2 \sin(\sqrt{\omega^2 - \lambda^2} t))$$

in the alternative form

$$x(t) = A e^{-\lambda t/2} \sin(\sqrt{\omega^2 - \lambda^2} t + \phi) \quad (23)$$

where $A = \sqrt{c_1^2 + c_2^2}$ and the phase angle ϕ is determined from the equations

$$\sin \phi = \frac{c_2}{A}, \quad \cos \phi = \frac{c_1}{A}, \quad \tan \phi = \frac{c_2}{c_1}$$

25) a) $F = 3.2 \text{ pounds}$ $\therefore m = \frac{w}{g}$

$$S = 1 \text{ ft} \quad = \frac{32}{32}$$

$$= \frac{1}{10} \text{ slug}$$

$\hookrightarrow F = k s$

$$\Rightarrow k = 2 \text{ lb/ft}$$

Here, $\beta = 0.4$

$$\therefore \lambda = \frac{\beta}{m} = \frac{0.4}{\frac{1}{10}} = 4 \quad \therefore \lambda = 2$$

$$\text{Now, } \lambda^2 - \omega^2$$

$$= 2^2 - (2\sqrt{5})^2 = -18$$

$$\therefore 2\lambda = \frac{B}{m} = \frac{0.4}{\frac{1}{10}} = 4 \quad \therefore \lambda = 2$$

$$\omega = \sqrt{\frac{k}{m}} = 2\sqrt{5}$$

→ underdamped

$$\therefore \text{The Equation as, } x(t) = e^{-\lambda t} (c_1 \cos \sqrt{\omega^2 - \lambda^2} t + c_2 \sin \sqrt{\omega^2 - \lambda^2} t)$$

$$= e^{-2t} (c_1 \cos 4t + c_2 \sin 4t)$$

Given Condition, $x(0) = -1$.

$$\therefore -1 = e^0 (c_1 \cos 0^\circ + c_2 \sin 0^\circ)$$

$$\Rightarrow c_1 = -1$$

Given Condition, since the mass was initially released, $x'(0) = 0$.

$$x'(t) = e^{-2t} (-4c_1 \sin 4t + 4c_2 \cos 4t) - 2e^{-2t} (c_1 \cos 4t + c_2 \sin 4t)$$

$$\Rightarrow 0 = e^0 (4 \sin 0 + 4c_2 \cos 0) - 2e^0 (-\cos 0 + c_2 \sin 0)$$

$$\Rightarrow 0 = 4c_2 + 2$$

$$\Rightarrow c_2 = -\frac{1}{2}$$

$$\Rightarrow c_2 = -\frac{1}{2}$$

$$\therefore x(t) = e^{-2t} \left(-\cos 4t - \frac{1}{2} \sin 4t \right)$$

b) Here, $A = \sqrt{(-1)^2 + \left(-\frac{1}{2}\right)^2} = \frac{\sqrt{5}}{2}$

$$\varphi = \tan^{-1} \left(\frac{-1}{-\frac{1}{2}} \right) = \tan^{-1}(2) = 1.1071 \quad (\text{in radian mode})$$

Assuming, $\alpha = 1.1071$.

$$\therefore x(t) = \frac{\sqrt{5}}{2} e^{-2t} \sin(4t + \alpha)$$

c) From the given condition,

$$x(t) = 0$$

$$\Rightarrow \frac{\sqrt{5}}{2} e^{-2t} \sin(4t + \alpha) = 0$$

$$\Rightarrow \sin(4t + \alpha) = 0$$

$$\Rightarrow 4t + \alpha = n\pi$$

$$\Rightarrow t = \frac{n\pi}{4} - \frac{\alpha}{4} \quad \text{where } n=0, 1, 2, 3, \dots$$

$$\Rightarrow t = \frac{n\pi}{4} - \frac{\pi}{4} \quad \text{where } n=0, 1, 2, 3, \dots$$

Here we have to find the value of n that satisfies the required conditions

if, $n=0 \rightarrow t = -0.2765$ (negative time not accepted)

if, $n=1 \rightarrow t = 0.508610$ but,

if, $x'(t) < 0$ then the mass heading upwards.

$$x'(t) = e^{2t}(-4c_1 \sin 4t + 4c_2 \cos 4t) - 2e^{2t}(c_1 \cos 4t + c_2 \sin 4t)$$

$$\hookrightarrow x'(0.508610) = 12.367 \text{ (+ve)}$$

if, $n=2 \rightarrow t = 1.294 \text{ sec}$

$$\hookrightarrow x'(1.294) = -59.492 \text{ (-ve)}$$

(-ve) $\Rightarrow$ up ward

(+ve) $\Rightarrow$ downward

$$\therefore t = 1.294$$

$\hookrightarrow$ passes through the equilibrium

first time upwards

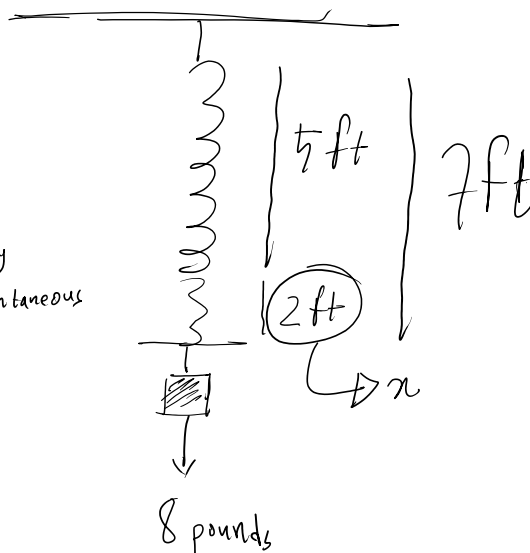
(26) (a)

10 pounds stretches 2 foot

8 " " 1.6 foot

$$\therefore m = \frac{8}{32} = 0.25 \text{ slugs. } x = 1.6 \text{ ft}$$

$$\begin{aligned} F &= kx \\ \Rightarrow k &= 5 \\ \beta &= 1 \\ \hookrightarrow &\text{as numerically} \\ &\text{equal to instantaneous} \\ &\text{velocity.} \end{aligned}$$



Nov. $\frac{d^2x}{dt^2} + \frac{\beta}{m} \frac{dx}{dt} + \frac{k}{m} x = 0$

$$\Rightarrow \frac{d^2x}{dt^2} + 4 \frac{dx}{dt} + 20x = 0$$

$$\hookrightarrow x(t) = e^{-2t} (C_1 \cos 4t + C_2 \sin 4t)$$

Given condition, $x(0) = \frac{1}{2} \text{ ft}$

$$\hookrightarrow C_1 = \frac{1}{2}$$

$$\begin{aligned} \omega^2 &= 20 \\ \Rightarrow \omega &= \sqrt{20} \\ &= 2\sqrt{5} \\ \frac{\beta}{m} &= 2\gamma \Rightarrow \gamma = 2 \\ \therefore \sqrt{\omega^2 - \gamma^2} &= 4 \\ \hookrightarrow &\text{Overdamped} \end{aligned}$$

$$\therefore x(t) = e^{-2t} \left(\frac{1}{2} \cos 4t + C_2 \sin 4t \right)$$

$$\begin{aligned} \Rightarrow x'(t) &= e^{-2t} (-2 \sin 4t + 4 C_2 \cos 4t) \\ &\quad - 2e^{-2t} \left(\frac{1}{2} \cos 4t + C_2 \sin 4t \right) \end{aligned}$$

$$\Rightarrow x'(0) = 1$$

$$\Rightarrow 1 = (0 + 4C_2) - 2\left(\frac{1}{2} + 0\right)$$

$$\Rightarrow C_2 = \frac{1}{2}$$

$$\therefore x(t) = e^{-2t} \left(\frac{1}{2} \cos 4t + \frac{1}{2} \sin 4t \right)$$

$$\textcircled{b} \quad A = \sqrt{c_1^2 + c_2^2} = \frac{1}{\sqrt{2}}$$

$$\varphi = \tan^{-1} \frac{c_1}{c_2} = \frac{\pi}{4}$$

$$x(t) = \frac{1}{\sqrt{2}} e^{-2t} \left(\sin 4t + \frac{\pi}{4} \right)$$

$\textcircled{c}$ (Similar as 2a(c))

Here $x'(t) < 0$ as it is downwards)

Answer: $t = \frac{7\pi}{16}$

Courtesy: Fattah 😊

$\textcircled{d}$ (try doing it by
using de Moivre)



you're using desmos) t

